SHREENIJA VADAYAR

PRODUCT DEVELOPMENT | DESIGN ENGINEER | PROTOTYPING & CREATIVE TECHNOLOGY

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PROFESSIONAL SUMMARY

Design and product-development engineer with hands-on experience taking concepts from CAD to working prototypes through SolidWorks, 3D printing, fabrication, and testing. Experienced collaborating across Engineering, Manufacturing, and Quality at Tecomet, with additional experience developing complex physical systems and interactive products. Lean Six Sigma Green Belt and First Prize winner of the Make It Wearable Engineering Student Contest, bringing a structured approach to problem-solving and iterative development.

TECHNICAL SKILLS

CAD & Design Engineering: SolidWorks | SketchUp | 3D Modeling | CAD Revisions | Engineering Drawings | Bills of Materials (BOMs) | Process Control Plans (PCPs) | Technical Documentation
Prototyping & Fabrication: 3D Printing | Laser Cutting | Fabrication File Preparation | Physical Prototyping | Assembly | Soldering | Wiring | Materials Evaluation | Iterative Testing
Electronics & Physical Computing: Arduino IDE | Arduino Uno | Adafruit Trinket M0 | VL53L0X Distance Sensors | HC-SR04 Ultrasonic Sensors | Micro Servos | NeoPixel LEDs | Piezo Buzzers | Bluetooth Connectivity
Programming & Interactive Technology: Python | Arduino Programming | Unity | Virtual Reality Development
Manufacturing & Quality: Visual Inspection | Dimensional Inspection | Tolerance Verification | Nonconformance Documentation | Process Control | Defect Investigation | Root-Cause Analysis | Continuous Improvement
Design & Productivity Tools: Adobe Illustrator | Adobe Photoshop | Figma | Microsoft Excel | Microsoft PowerPoint | Google Workspace

EXPERIENCE

TECOMET — Engineering Intern / Co-Op | Boulder, CO | ***June 2022 - July 2023***

- Supported medical-device component development by translating evolving client requirements into revisions of existing SolidWorks models, engineering drawings, Bills of Materials (BOMs), and Process Control Plans (PCPs).
- Coordinated engineering documentation and design updates across Engineering, Manufacturing, and Quality, helping maintain alignment between client specifications, production requirements, and project deliverables.
- Performed visual and dimensional inspections of manufactured components, evaluating surface defects and measurements against engineering drawings, specifications, and tolerances.
- Documented nonconformances and communicated recurring defects to departmental leads, contributing inspection findings and production data to cross-functional investigations.
- Supported a process-improvement initiative addressing recurring dimensional and visual defects by expanding inspection samples, collecting and analyzing data, and collaborating with Engineering, Manufacturing, and Quality to evaluate potential causes and solutions.
- Contributed to implementation of a revised production-lot size that substantially reduced defects, then monitored subsequent lots for approximately two weeks to help confirm that improvements were sustained.
- Prepared for client meetings, documented meeting notes and decisions, communicated design revisions, and followed up on action items to support cross-functional coordination and project progress.
- Revised and maintained 50+ engineering documents, including SolidWorks models, engineering drawings, Bills of Materials (BOMs), and Process Control Plans (PCPs), supporting 20–25+ parts orders and coordinating 2–3 concurrent workstreams.

ATLAS INSTITUTE, UNIVERSITY OF COLORADO BOULDER — Undergraduate Research Assistant | Boulder, CO | *January 2023 - May 2023*

- Evaluated fabrication methods and material options for a graduate research project through hands-on prototyping, structured testing, and comparative analysis.
- Produced and evaluated 20+ prototype samples and trials using 3D printing, laser cutting, and alternative materials, including filaments and textiles used for component casing and cushioning.
- Used SketchUp and SolidWorks to support 3D modeling and prototype revisions, assessing fabrication quality, dimensional precision, production time, material behavior, and physical characteristics.
- Recorded experimental observations and analyzed results in Microsoft Excel and through manual review to compare prototype performance and fabrication outcomes.
- Documented findings and communicated results to undergraduate and graduate researchers, informing subsequent experiments and material and fabrication decisions for a final prototype prepared for a funding presentation.

SELECTED PROJECTS

WOVEN VOICES — Interactive Wearable Technology Installation | Senior Capstone | Lead Designer & Developer | Solo Project | *September 2024 - May 2025*

- Designed and fabricated a wearable installation featuring seven mechanical flowers, combining garment construction, custom mechanisms, brass components, 3D-printed housings, electronics, and interactive lighting.
- Modeled and fabricated flower components and sensor housings, integrating brass-rod mechanisms, micro servos, VL53L0X distance sensors, Adafruit Trinket M0 boards, and NeoPixel LEDs.
- Developed Arduino programs for individual flowers to respond to visitor proximity by adjusting servo movement and LED brightness; programmed additional lighting effects for the garment.
- Iteratively tested, assembled, and troubleshot mechanical, electronic, and textile components, addressing challenges involving servo mounting, structural rigidity, wiring, power, and system integration.
- Presented the interactive wearable at the 2025 CU Boulder Design Expo, engaging directly with more than 25 visitors and demonstrating the installation's proximity-responsive behavior.
- **Recognition: First Prize — Autodesk Make it Wearable Engineering Student Contest.**

VIRTUAL REALITY DRAWING ENVIRONMENT — Interactive 3D Experience | University of Colorado Boulder

- Independently developed a virtual environment from scratch in Unity for a Meta VR headset, creating an interactive space for open-ended drawing and creative exploration.
- Implemented interactive drawing functionality, color selection through a palette and color wheel, and erasing tools to support user-directed creation.
- Designed virtual drawing surfaces and objects, including mannequins and a drawing board, to give users multiple ways to create and interact within the environment.
- Used Python programming as part of the project development process within the Unity environment.

TAILGATING DETECTION DEVICE — Physical Computing & Object Design | University of Colorado Boulder

- Independently designed, programmed, and fabricated a functional detection prototype integrating an Arduino Uno, HC-SR04 ultrasonic distance sensor, piezo buzzer, and NeoPixel LEDs.
- Developed a laser-cut enclosure and prepared fabrication artwork using Adobe Illustrator.
- Programmed and tested sensor-triggered alerts and LED behavior, including scrolling messages and visual feedback.
- Evaluated sensor detection behavior and system functionality, identifying the ultrasonic sensor's forward-facing detection range as a design constraint.

GUARDIAN ANGEL — Wearable Safety Technology | University of Colorado Boulder | Team Project

- Designed the mechanical form of a wearable safety button as part of a four-person team developing a connected safety concept.
- Contributed to electronics integration, Arduino programming, and testing Bluetooth connectivity between the wearable device and its companion experience.
- Designed and developed elements of the companion interface in Figma, supporting the connection between the physical product and its digital interaction.

MOSSTAK — Sustainable Living Roof Concept | University of Colorado Boulder | Team Project

- Contributed to a conceptual sustainable roofing system incorporating moss-tile design, using SketchUp modeling and 3D printing to create a physical representation of the proposed product.
- Conducted market and sustainability research to inform the concept and its potential value to users.
- Developed UX/UI flows in Figma and contributed to website and pitch-presentation materials to communicate the product concept.

SOLEMATE — Wearable Running Analysis Product Concept | University of Colorado Boulder | Team Project

- Contributed engineering research to a wearable running-analysis concept, investigating pressure-sensor placement, sensor quantity, sole configuration, and product cost considerations.
- Conducted user interviews, surveys, competitor research, and market analysis to understand runner needs and inform concept development.
- Evaluated potential product configurations and technical trade-offs for a concept intended to help runners better understand their running mechanics without relying exclusively on personal training services.

EDUCATION

University of Colorado Boulder — B.S. Creative Technology & Design | Minor in Business | Minor in Leadership

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

LEAN SIX SIGMA GREEN BELT — *Certified Green Belt*

• Council for Six Sigma Certification (CSSC) | July 2026

WOMEN'S STARTUP PROGRAM — MANAGEMENT & ENTREPRENEURSHIP CERTIFICATE — *IIM Bangalore (NSRCEL) & Goldman Sachs*

• Entrepreneurship, product validation, business planning, pitching, and business development